Your grade: 70%

Your latest: **70%** • Your highest: **70%** • To pass you need at least 80%. We keep your highest score.

Retry

1. **A bicycle rental company has counted the number of bicycle rentals in each season (spring, summer, fall, winter) for the past two years.**

Additionally, the company has collected weather data (temperature, wind speed and humidity).

Use the data for bicycle rentals and weather presented in the tables and graphs below to answer these practice quiz questions.

1 / 1 point

Which proportion describes the most popular season for renting bicycles in Year 1?

	Year 1	Year 2
Spring	150,000	321,348
Summer	347,316	571,273
Fall	419,650	641,479
Winter	326,137	515,476
Total	1,243,103	2,049,576

- 150,000 / 1,243,103
- 641,479 / 2,049,576
- 419,650 / 1,243,103

  Correct!

- 1,861,129 / 3,292,679

2. Which proportion describes the least popular season for renting bicycles in Year 2?

1 / 1 point

	Year 1	Year 2	Total
Spring	150,000	321,348	471,348
Summer	347,316	571,273	918,589
Fall	419,650	641,479	1,061,129
Winter	326,137	515,476	841,613
Total	1,243,103	2,049,576	3,292,679

- 2,049,576 / 3,292,679
- 321,348 / 471,348
- 641,479 / 2,049,576
- 471,348 / 3,292,679
- 321,348 / 2,049,576

  Correct!

3. Which statement best describes the meaning of 326,137 / 841,613?

1 point

	Year 1	Year 2	Total
Spring	150,000	321,348	471,348
Summer	347,316	571,273	918,589
Fall	419,650	641,479	1,061,129
Winter	**326,137**	515,476	**841,613**
Total	1,243,103	2,049,576	3,292,679

- The proportion of Total Winter rentals that occurred in Year 1.
- The proportion of Total rentals that occurred in Year 1.
- The proportion of Total rentals that occurred in Winter.
- The proportion of Year 1 rentals that occurred in Winter.

  Not quite! That would be 326,137 / 1,243,103.

4. How do the proportion of rides in the Summer compare between Year 1 and Year 2?

1 point

	Year 1	Year 2	Total
Spring	150,000	321,348	471,348
Summer	347,316	571,273	918,589
Fall	419,650	641,479	1,061,129
Winter	326,137	515,476	841,613
Total	1,243,103	2,049,576	3,292,679

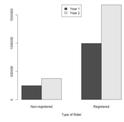
- The proportion is higher in Year 2 because 571,273 is larger than 347,316.

  Not quite! Proportions are not found using only the number of rides in the Summer from both years. What other number do we need to look at?

- The proportion is higher in Year 1 because 2,049,576 is larger than 1,243,103.
- Can't tell without doing additional calculations.

5. The company suspects that they will have a larger increase in rentals from registered riders compared to non-registered riders over the two years. They make this visualization to see how the numbers compare. For which group does the increase in riders seem larger?

1 / 1 point



Data from visualization:

Rider Type	Year 1	Year 2
Non-registered	250,000	400,000
Registered	1,000,000	1,700,000

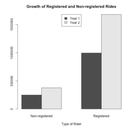
- Registered

  Good job! The increase in the height of the registered rides from Year 1 to Year 2 does seem to be the largest.

- Can't tell from this graph
- Non-registered
- They look the same

6. What kind of graph is this?

1 / 1 point



Beneath the title, there are two labeled categories along the bottom: "Non-registered" on the left and "Registered" on the right.

Each category contains two rectangular shapes: one dark and one light. The dark shapes are labeled "Year 1" in a small legend, and the light shapes are labeled "Year 2."

The vertical axis on the left side is marked with numbers increasing from 0 at the bottom up to 1,500,000 near the top.

- Bar chart
- Side-by-side bar chart

  Good job! There are groupings of bars within this graph. This indicates that there are at least two groups being compared in this side-by-side bar chart.

- Stacked bar chart
- Mosaic plot

7. The bicycle company is interested in knowing how rides are affected by various weather conditions. To start with, they want to examine the registered wind speeds (after a normalization).

1 / 1 point

Is wind speed a discrete or continuous variable?

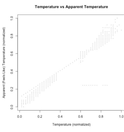
- Discrete
- Continuous

  Good job! Wind speed is a quantitative continuous variable.

- Can't tell

8. The company wants to consider how weather patterns affect the bicycle rentals. They first consider how the measured temperature compares to the apparent temperature, or the temperature that humans perceive it to be. The temperatures have been normalized to fall on a scale between 0 and 1. Yesterday the normalized real temperature was 0.4. Today the normalized real temperature is 0.6. Which day would you expect to have a higher apparent temperature?

1 / 1 point



The image has the title "*Temperature vs Apparent Temperature.*"

It shows a plot with:

- A horizontal axis at the bottom labeled "*Temperature (normalized)*" ranging from 0.0 to 1.0.
- A vertical axis on the left labeled "*Apparent Feels-Like Temperature (normalized)*" also ranging from 0.0 to 1.0.

Inside the plot are many small black dots. Most of the dots form a diagonal band stretching from the bottom-left corner (near 0,0) up to the top-right corner (near 1,1). There are also some dots that sit in a horizontal line just above 0.2 on the vertical axis, toward the right side of the plot between 0.6 and 0.9 on the horizontal axis.

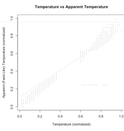
- Yesterday
- Today

  Good job! Because the scatterplot has a positive direction, we expect the apparent temperature to be larger when the actual temperature is larger.

- Can't tell

9. The scatterplot between temperature and apparent temperature is linear. What is the strength of the scatterplot between temperature and apparent temperature?

1 / 1 point



The image has the title "*Temperature vs Apparent Temperature.*"

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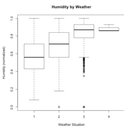
- Strong

  Good job! Almost all of the point are very close to a line, so we consider the strength of the linear relationship to be strong.

- Moderate
- Weak
- Can't tell

10. Eventually, the bicycle company wants to think about how bicycle rides vary based on weather. After looking at humidity, they think that the humidity might be associated with the general weather conditions. They consider weather situations of 1 = clear to partly cloudy, 2 = misty with no to some clouds, 3 = light rain and light snow, and 4 = heavy rain, snow, thunderstorms, and other extreme weather. Based on the side-by-side boxplots below, which weather condition has the highest mean humidity?

1 point



Summary:

Weather Situation	lower whisker end	lower box end	box middle line	upper box end	upper whisker end	data beyond whiskers
1	0.10	0.45	0.55	0.70	1.00	None
2	0.20	0.55	0.70	0.85	1.00	None
3	0.55	0.80	0.86	0.92	1.00	Many values from 0.35 to 0.55
4	0.85	0.85	0.85	0.90	0.92	None

- 1 = Clear to partly cloudy
- 2 = Misty with no to some clouds
- 3 = Light rain and light snow

  Not quite! Think about how a mean is represented in boxplots.

- 4 = Heavy rain, snow, thunderstorms, and other extreme weather
- Can't tell